Whole-exome sequencing in the evaluation of fetal structural anomalies: a prospective cohort study

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Summary

Background **Identification of chromosomal aneuploidies and copy number variants that are associated with fetal structural anomalies has substantial value. Although whole-exome sequencing (WES) has been applied to case series of a few selected prenatal cases, its value in routine clinical settings has not been prospectively assessed in a large unselected cohort of fetuses with structural anomalies. We therefore aimed to determine the incremental diagnostic yield (ie, the added value) of WES following uninformative results of standard investigations with karyotype testing and chromosomal microarray in an unselected cohort of sequential pregnancies showing fetal structural anomalies.**

Methods **In this prospective cohort study, the parents of fetuses who were found to have a structural anomaly in a prenatal ultrasound were screened for possible participation in the study. These participants were predominantly identified in or were referred to the Columbia University Carmen and John Thain Center for Prenatal Pediatrics (New York, NY, USA). Fetuses with confirmed aneuploidy or a causal pathogenic copy number variant were excluded from WES analyses. By use of WES of the fetuses and parents (parent–fetus trios), we identified genetic variants that indicated an underlying cause (diagnostic genetic variants) and genetic variants that met the criteria of bioinformatic signatures that had previously been described to be significantly enriched among diagnostic genetic variants.**

Findings **Between April 24, 2015, and April 19, 2017, 517 sequentially identified pregnant women found to have fetuses with a structural anomaly were screened for their eligibility for inclusion in our study. 71 (14%) couples declined testing, 87 (17%) trios were missing at least one DNA sample (from either parent or the fetus), 69 (13%) trios had a clinically relevant abnormal karyotype or chromosomal microarray finding, 51 (10%) couples did not consent to WES or withdrew consent, and five (1%) samples were not of good enough quality for analysis. DNA samples from 234 (45%) eligible trios were therefore used for analysis of the primary outcome. By use of trio sequence data, we identified diagnostic genetic variants in 24 (10%) families. Mutations with bioinformatic signatures that were indicative of pathogenicity but with insufficient evidence to be considered diagnostic were also evaluated; 46 (20%) of the 234 fetuses assessed were found to have such signatures.**

Interpretation **Our analysis of WES data in a prospective cohort of unselected fetuses with structural anomalies shows the value added by WES following the use of routine genetic tests. Our findings suggest that, in cases of fetal anomalies in which assessment with karyotype testing and chromosomal microarray fail to determine the underlying cause of a structural anomaly, WES can add clinically relevant information that could assist current management of a pregnancy. The unique challenges of WES-based prenatal diagnostics require analysis by a multidisciplinary team of perinatal practitioners and laboratory specialists.**

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Introduction

With the introduction of ultrasound into obstetrical care, the identification of fetal structural anomalies has become routine. When anomalies are found, further evaluation frequently includes karyotype testing to detect whole chromosome aneuploidies and chromosomal microarray analysis (CMA) to identify smaller microdeletions and duplications, which are collectively referred to as copy number variants (CNVs). Overall, approximately 32% of fetuses with a structural anomaly identified by ultrasound have a clinically relevant abnormal karyotype, and an additional 6% have a causative CNV,[1] which means that more than 60% of these pregnancies do not have a definitive genomic diagnosis to guide future care and genetic counselling.

Whole-exome sequencing (WES) yields a diagnosis of the underlying genetic cause in 25–35% of children with an unexplained presumed genetic disorder (such as a birth defect) after negative findings in karyotype testing and chromosomal microarray.[2–4] This result in children suggests that WES could be an important diagnostic approach in understanding the underlying genetic causes of fetal structural anomalies, and this approach could lead to improved perinatal care and prognostic

Research in context

Evidence before this study

We searched PubMed for studies that used whole-exome sequencing (WES) on prenatal samples in humans with the search terms "prenatal diagnosis", "whole exome sequencing", congenital anomalies", and "amniocentesis". We only included publications if they described genetic testing by use of WES in prenatal cases of fetal anomalies that had been identified in utero, and we included cases that ended in stillbirths or pregnancy terminations. We only included studies that were published in English between Jan 1, 2012, and Jan 1, 2018, for which the full text of the article was available. We excluded studies with fewer than five cases of fetal structural anomalies. We included proband-only and trio sequencing, and not all studies reported a normal karyotype or chromosomal microarray before proceeding to WES. Overall, there were eight studies that met our inclusion criteria, which comprised 246 cases. Only two of these studies included more than 50 cases (84 cases in one study and 51 cases in the other). The two largest studies were restricted to pregnancies that resulted in stillbirths and pregnancy terminations and consisted of highly selected cases. The diagnostic rate ranged from 10% to 32·8% in studies including 20 or more fetuses, and a higher detection rate was noted for fetuses with multiple anomalies. These findings, although limited, suggest that prenatal WES could be of benefit in the evaluation of fetuses with structural anomalies. Identification of a specific diagnosis would allow more focused prenatal and neonatal treatment and improve parental genetic counselling. However, the present literature predominantly includes small studies of fetuses that have been genetically sequenced because of a high a priori suspicion of a genetic disease. This usage is not reflective of how WES is likely to be implemented in practice when introduced into clinical care, and patient selection criteria are likely to be broader in clinical use. Increased knowledge and experience is needed to determine how best to introduce WES into practice.

Added value of this study

To our knowledge, this study is the largest to date to evaluate the use of WES in the assessment of fetal anomalies. We report on consecutive pregnancies in which fetal structural anomalies were detected by routine ultrasound analysis, and WES was offered to all patients, regardless of the anomaly. We detected diagnostic genetic variants in 10% of fetuses and in 19% of fetuses with more than one anomaly. The frequency of genetic diagnoses varied by the organ system involved; lymphatic or effusion (24% of fetuses that were found to have diagnostic genetic anomalies), skeletal (24% of fetuses), central nervous system (22% of fetuses), and renal (16% of fetuses) anomalies were associated with the highest frequency of diagnostic genetic variants, and cardiac anomalies (5% of fetuses) were associated with the lowest frequency of diagnostic genetic variants. In addition to confirmed genetic diagnoses, we demonstrated a high frequency of bioinformatic signatures that were suggestive of a genetic cause but that do not presently meet diagnostic criteria, which shows that, with increasing experience and accumulated knowledge, the frequency of structural anomalies with diagnostic genetic variants by WES could increase.

Implications of all the available evidence

Evidence to date demonstrates the feasibility of WES for prenatal diagnosis and its value beyond karyotyping and chromosomal microarray in the assessment of a fetus with a structural anomaly. Presently, more than half of fetal structural anomalies have no proven cause. WES can substantially increase understanding of the underlying genetic mechanisms, to improve prenatal and transitional perinatal care. The knowledge gained from WES will improve prognostic counselling, leading to informed parental decision making. Based on our findings, it is likely that WES will transition into care shortly. However, practitioners should be aware of the limitations and potential difficulties in interpreting fetal genomic variants and of the need for a multidisciplinary team of molecular and clinical genetic experts and fetal imaging specialists.

parental genetic counselling. To date, only case series of a few highly selected fetuses (eg, in those with ultrasound findings consistent with a known genetic diagnosis) have evaluated the feasibility and value of WES in prenatal diagnostic testing; overall, only eight case series of 246 pregnancies have enrolled ten fetuses or more.[5–11] Of these, only two studies[7,11] reported on 50 cases or more. These studies restricted the participants to stillbirths, fetuses in terminated pregnancies, and other highly selected cases. The proportion of specific diagnostic genetic variants that were detected ranged from 10% to 32·8% in case series that included 20 fetuses or more.[5,11] Despite these limited data, laboratories and clinicians have begun to offer prenatal WES as part of clinical care in the evaluation of fetal anomalies. To guide this impending change to clinical care, we aimed to determine the number of diagnostic genetic variants identified by WES following the use of standard investigations with karyotype testing and CMA, in an unselected cohort of sequential pregnancies showing fetal structural anomalies.

Methods

Study design and participants

In this prospective cohort study in one tertiary level referral centre (Columbia University Carmen and John Thain Center for Prenatal Pediatrics, New York, NY, USA), the parents of fetuses that were found to have a structural anomaly during a prenatal ultrasound (at 11–35 weeks of pregnancy) were invited to participate. Couples were scanned by maternal fetal medicine physicians with additional training and expertise in fetal

imaging. Those with any structural anomaly were invited to participate by the study genetic counsellors (JLG, MS, ES, KB). All fetal structural anomalies, including nuchal translucency of more than 3·5 mm, were included. We also required that all participants had singleton pregnancies and were able to provide a fetal sample. Fetuses with a known infection or exposure to a known teratogenic drug, families with a known diagnosis of a genetic disorder, and cases in which a parental DNA sample was not available were excluded. Fetuses with ultrasound soft markers that were suggestive of Down syndrome but that showed no other anomalies, those with an isolated nuchal translucency of less than 3·5 mm, and those with abnormal karyotype or CMA results that were considered causative of the anomaly were also excluded. Written informed consent was obtained by a study-certified genetic counsellor. The institutional review board of Columbia University College of Physicians and Surgeons (New York, NY, USA) approved our study.

Procedures

The pregnant women received prenatal testing for karyotype and CMA; this testing included chorionic villus sampling, umbilical cord blood sampling, or amniocentesis as part of their clinical care. Fetal pleural effusions were assessed with fetal thoracocentesis and, in these cases, an aliquot of pleural fluid was used for WES. Parental blood samples were obtained at the time of the prenatal diagnostic procedure. An aliquot of DNA was isolated by the clinical laboratory from both the fetal and parental samples, which was de-identified and sent to the Institute for Genomic Medicine (Columbia University Irving Medical Center, New York, NY, USA) for WES. WES was not done if the karyotype or CMA results were abnormal and were considered causative of the anomaly.

If prenatal diagnostic testing was not done, the fetus, mother, and father (referred to as the parent–fetus trio) were enrolled in the study by the study genetic counsellors (JLG, MS, ES, KB) at the time of ultrasound diagnosis of the fetal structural anomaly and parental blood samples were obtained and stored for later analysis. At delivery, cord blood or fetal tissue (namely a section from the placenta or umbilical cord) were obtained and were assessed with a karyotype test, CMA, and WES. In cases of fetal demise or pregnancy termination, fetal tissue samples were used for analyses. If an immediate birth sample was unavailable, a neonatal sample was obtained within 72 h of delivery. At the time of enrolment, parents were informed that sequencing was for research purposes, only causative genetic variants would be reported, and results would take 4–8 weeks to be returned.

DNA samples from parent–fetus trios were exome sequenced by the Institute for Genomic Medicine, Columbia University Medical Center, with the Illumina HiSeq 2500 platform and KAPA Biosystem's library preparation kits, and whole-exome capture was done with Nimblegen SeqCap EZ version 3.0 rapid or Nimblegen SeqCap EZ version 4. Bioinformatic processing was done by the Institute for Genomic Medicine's bioinformatic and trio analysis platforms, which identified qualifying genotypes after accounting for the parental genotypes and genotypes present in in-house and external control reference samples (appendix).[3,12,13]

See Online for appendix

We used parent–fetus trio WES and an updated version of our established trio analysis framework, which has been used in several studies[3,12,13] of undiagnosed and unexplained genetic diseases, to allow rapid and efficient identification of putative de novo and inherited variants. Our bioinformatics workflow is shown in the appendix; in our analysis, we focus on two tiers of qualifying genotypes. Tier 1 was associated with the assumption that a relevant genotype will be highly penetrant and is expected to be absent in the parents and the available internal and external controls. Tier 1 genotypes also have higher variant calling quality requirements (appendix). Tier 2 is a literature-motivated screen, which permitted the genotypes to be observed at low frequencies among controls (internal and external) and had more relaxed genotype requirements (appendix); however, to qualify as tier 2, the genotype either had to have previously been classified as likely pathogenic by ClinVar or the Human Gene Mutation Database or had to be a protein truncating variant that affected a known disease-causative gene through protein-truncating genotypes. Variants that qualify within these two tiers were evaluated by a multidisciplinary team of researchers and clinical investigators. During the study, qualifying variants were periodically re-assessed and linked to the latest ClinVar, ClinGen, Human Gene Mutation Database, and Online Mendelian Inheritance in Man literature.[14] When new literature supported a possible clinical association, the variants were re-evaluated.

For **ClinVar** see www.ncbi.nlm.nih.gov/clinvar

For the **Human Gene Mutation Database** see www.hgmd.cf.ac.uk

Every qualifying variant was considered in the context of the American College of Medical Genetics and Genomics recommendations, including whether the genotype had been reported as pathogenic in other patients or whether the genotype was of the same class observed among previous cases of the relevant American College of Medical Genetics and Genomics indication (eg, protein-truncating variants of a gene when loss-of-function variants in that gene had been reported to be pathogenic).[15] For each of the qualifying genotypes, we used the Online Mendelian Inheritance in Man database to check for existing disease associations that included at least partial overlap with the fetal phenotype. We also required that the inheritance model was consistent with that of reported cases (ie, dominant or recessive).

The qualifying variants were presented to a multidisciplinary conference that comprised clinical and molecular geneticists, population geneticists, genetic counsellors, maternal fetal medicine specialists, neonatologists, and imaging experts, to determine relevance to the clinical presentation. When the team confirmed a

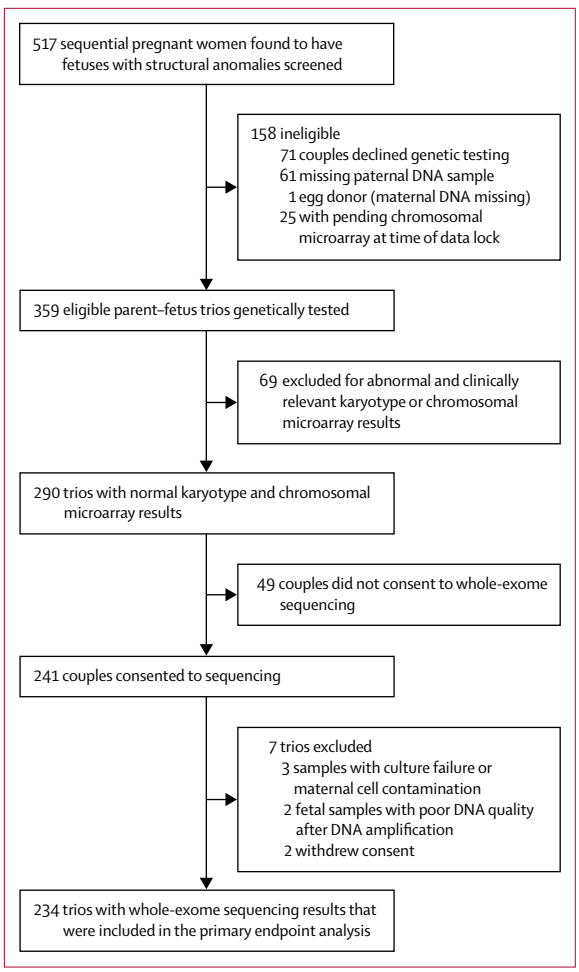

Figure 1: **Study profile**
Abnormal karyotype and chromosomal microarray findings that led to exclusions are shown in the appendix.

genotype–phenotype match, DNA samples from the trio were Sanger sequenced in our clinical molecular laboratory. Following confirmation of the variant by Sanger sequencing, these fetuses were considered to have a genetic diagnosis and the findings were reported to the parents. Incidental findings from the American College of Medical Genetics and Genomics recommended list[15,16] were not routinely evaluated and were only reported to the parents if they emerged through our trio analysis framework (appendix).

Although the genetic diagnoses emerging from the tier 1 and tier 2 screens followed American College of Medical Genetics and Genomics guidelines, we also used previously published bioinformatic approaches[2,3,17] to estimate how many qualifying genotypes were not classified as diagnostic because of inadequate knowledge of anomaly-associated genes or other factors. These bioinformatic signatures were insufficient to confirm pathogenicity, but they have been shown to be systematically increased in cohorts of patients with well documented genetic diseases and can therefore serve as a useful approximation of the number of likely causative mutations that were not confirmed (appendix).

Outcomes

The primary endpoint, which was assessed in all parent–fetus trios with available WES data, was the detection of diagnostic genetic variants that caused the fetal developmental anomaly. Secondary endpoints were a comparison of the frequency of fetuses with a diagnostic genetic variant when comparing cultured versus direct sampling and single versus multiple system anomalies. We also assessed the exploratory endpoint of the presence of bioinformatic signatures that were often associated with and likely causative of the fetal structural anomaly. All endpoint statistical analyses were done with R (version 3.4.1).

Role of the funding source

The funder of the study was involved at all stages because the study was funded internally with no outside support.. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

Between April 24, 2015, and April 19, 2017, 517 sequentially identified pregnant women who were found to have fetuses with structural anomalies were screened for their eligibility for inclusion in our study (figure 1; appendix). 71 (14%) couples declined testing, 87 (17%) trios were missing at least one DNA sample (from either parent or the fetus), 69 (13%) trios had a clinically relevant abnormal karyotype or CMA finding, 51 (10%) couples did not consent to WES or withdrew consent, and five (1%) samples were not of good enough quality for analysis. DNA samples from 234 (45%) eligible trios were therefore used for analysis of the primary outcome.

These 234 trios were ethnically diverse: in 171 (73·1%) trios, parents self-reported an ancestry other than European white or sequence data placed the fetus outside principal component clusters indicative of the European White population (appendix). This representation is in marked contrast with what has been seen in other genomics studies, which had a predominantly European white population.[18] An exome-based relatedness screen of all parental pairs identified three parental pairs to be related up to the third degree, and these three pairs represented the only known consanguineous families (appendix).

Fetal DNA was obtained from amniocentesis for 99 (42%) fetuses, chorionic villus sampling for 57 (24%) fetuses, cord blood at birth for 30 (13%) fetuses, tissue samples after miscarriage or pregnancy termination for 26 (11%) fetuses, cord segment at birth for seven (3%) fetuses, neonatal buccal swabs for

six (3%) babies, percutaneous umbilical fetal blood sampling for three (1%) fetuses, sections of the placenta for two (1%) fetuses, neonatal blood for two (1%) fetuses, and pleural fluid for two (1%) fetuses. 82 (35%) fetal DNA samples were sequenced from cultured samples. 134 (57%) pregnancies resulted in livebirths, 78 (33%) pregnancies were terminated, eight (3%) fetuses were stillborn or died in utero, and no information about the pregnancy outcome was available for 14 (6%) fetuses.

The exome-wide average read coverage was 89·3 (SD 22·6) reads and the per sample coverage statistics are shown in the appendix. Per fetal sample, we achieved at least ten sequence reads of coverage across 96·1% (1·7%) of the 33·8 megabase pairs of the consensus coding sequence (Consensus CDS project, release 20).

We identified 1122 tier 1 qualifying genotypes, with an average of 4·8 (SD 3·6) qualifying genotypes per fetus. These were distributed across four inheritance models (appendix): (1) 258 putative de novo mutations (average 1·1 [1·1] per fetus; figure 2); (2) 130 X-chromosome hemizygous genotypes (average 1·0 [1·0] per male fetus); (3) 80 homozygous genotypes (average 0·34 [1·4] per fetus; 0·20 [0·5] after removal of the three consanguineous families); and (4) 654 compound heterozygous genotypes (average 2·8 [3·0] per fetus). Consistent with our previous findings,[19] the prevalence of inherited qualifying genotypes were significantly associated with genetic ancestry (Mann-Whitney test; p<0·0001; appendix). As more comprehensive and ethnically diverse population reference cohorts become available, this effect will reduce; however, at present this association affects interpretation. Overall, the prevalence of qualifying variants did not significantly differ between direct and cultured samples (Mann-Whitney test; p=0·6).

Across the 234 trios, we found 60 tier 2 hemizygous, homozygous, or putative de novo qualifying genotypes (average 0·26 [SD 0·57] per fetus). A tier 2 compound heterozygous screen identified an additional 156 genotypes (average 0·67 [0·98] per fetus) in which at least one of the two variants that contributed to a compound heterozygous genotype qualified for the tier 2 criteria (appendix). We also evaluated 38 fetal genotypes that were protein truncating or pathogenic-reported genotypes or both in the medical literature.

After consensus agreement among the investigators, we found that 24 (10%) of 234 fetuses had a diagnostic genetic variant (ie, WES-based genetic diagnosis; table 1). 15 (63%) variants were de novo mutations, six (25%) variants were from recessive genotypes (including a male hemizygous *L1CAM* genotype, homozygous *TMEM67, ARMC9*, and *C5ORF42* genotypes, and compound heterozygous *FLVCR2* and *RAPSN* bi-allelic loss-of-function genotypes), and three (13%) variants were inherited in an autosomal dominant way from a previously undiagnosed parent (*PKD2, RASA1*, and *SOS1*; appendix). In the case of the fetus with the inherited *PKD2* frameshift variant, the transmitting mother was unaware of her diagnosis but was subsequently found to have polycystic kidneys (table 1). The mother transmitting *RASA1* had no clinical imaging studies and thus remains undiagnosed, and the transmitting father with *SOS1* had hydrops as a fetus that he only became aware of after the birth of his affected child (appendix).

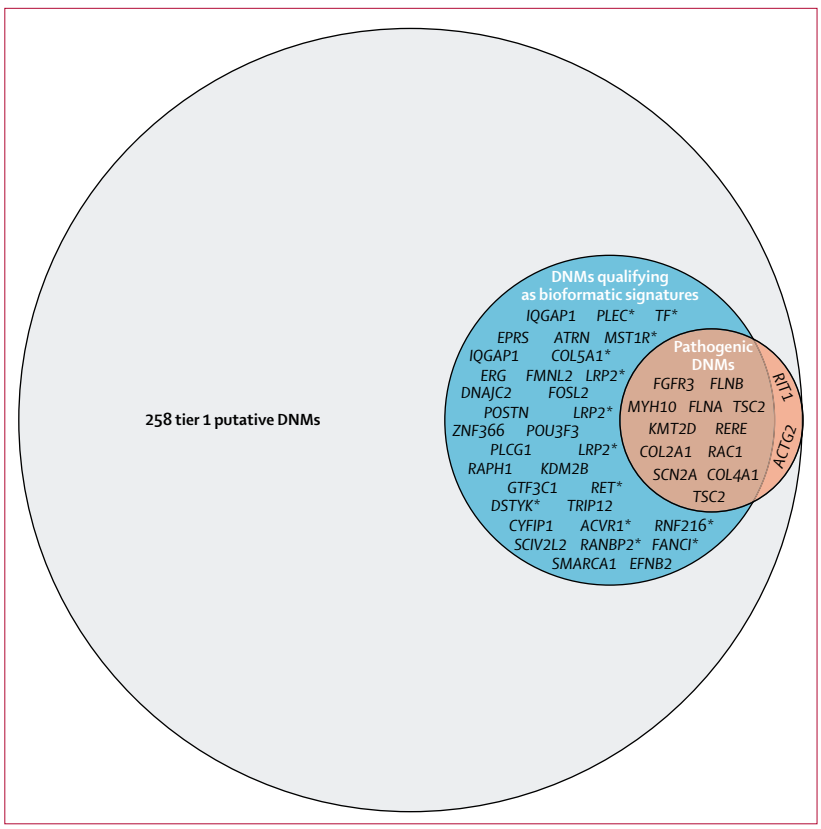

Figure 2: **Area-proportional Venn diagram of the overlap between tier 1 putative DNMs, DNMs that represented bioinformatic signatures, and pathogenic tier 1 DNMs that formed the diagnostic genetic variants**

DNMs=de novo mutations. *DNMs determined to affect disease-associated genes on the basis of a search in the Online Mendelian Inheritance in Man database in July, 2017.

For the **Consensus CDS project** see www.ncbi.nlm.nih.gov/ccds

We found that 46 (20%) of 234 fetuses had mutations meeting our bioinformatic criteria (appendix; ie, bioinformatic signatures). These variants were judged to have substantial potential of being pathogenic but did not meet our criteria to be confirmed as causative of the structural anomaly. Under American College of Medical Genetics and Genomics guidelines,[15,16] 15 (33%) of these 46 bioinformatic signatures would be considered pathogenic. In agreement with other reported diseases, the prevalence of such bioinformatic signatures in our cohort was substantially greater than that found in unaffected populations.[2,3,17]

Among the 160 fetuses with one anomaly, ten (6%) fetuses were found to have a diagnostic genetic variant, whereas 14 (19%) fetuses with several anomalies had a diagnostic genetic variant (Fisher's exact two-tailed test; odds ratio 3·5, 95% CI 1·4–9·3; p=0·005). A greater proportion of diagnostic genetic variants were identified

	Prenatal imaging phenotype	Gene	DNA alteration*	cDNA alteration, protein alteration	RefSeq identifier	Consequence	Inheritance	Novel or previously reported (PMID)
fetal0023F	Subependymal nodules, representing periventricular nodular heterotopia; prominent cisterna magna	*FLNA*	Chr X, 153582554, TC→T	5521delG, Glu1841ArgfsX39	NM_001110556.1	Frameshift	De novo	Novel
fetal0056F†	Bilateral ventriculomegaly, acqueductal stenosis	*MYH10*	Chr 17, 8417118, G→A	2599C→T, Arg867X	NM_001256012.1	Nonsense	De novo	Novel
fetal0126F	Rhabdomyomas, subependymal nodules	*TSC2*	Chr 16, 2120571, C→T	1831C→T, Arg611Trp	NM_000548.4	Missense	De novo	Tuberous sclerosis (8824881)
fetal0004F	Agenesis of the corpus callosum	*L1CAM*	Chr X, 153134145, G→A	1417C→T, Arg473Cys	NM_000425.4	Missense	Hemizygous male fetus	L1 syndrome (9744477)
fetal0116F	Cystic hygroma, hydrops, hydranencephaly, arthrogryposis	*FLVCR2*	Chr 14, 76108242, G→A	1509+1G→A, NA	NM_017791.2	Splice donor	Compound heterozygous	Novel
fetal0116F	Cystic hygroma, hydrops, hydranencephaly, arthrogryposis	*FLVCR2*	Chr 14, 76100019, A→AT	1001dupT, Met334IlefsX37	NM_017791.2	Frameshift	Compound heterozygous	Novel
fetal0166F	Splaying of cerebellar hemispheres, bilateral multicystic kidneys	*TMEM67*	Chr 8, 94777801, CAG→C	579_580delAG, Gly195IlefsX13	NM_153704.5	Frameshift	Homozygous	Meckelin-related ciliopathy (20232449)
fetal0108F†	Abnormal profile, cleft palate, encephalocele, holoprosencephaly, polydactyly	*ARMC9*	Chr 2, 232091490, G→T	601G→T, Glu201Ter	NM_025139.5	Nonsense	Homozygous	Novel
fetal0026F	Severe skeletal dysplasia	*FGFR3*	Chr 4, 1806099, A→G	1124A→G, Tyr375Cys	NM_001163213.1	Missense	De novo	Thanatophoric dysplasia (8845844)
fetal0165F	Micrognathia, micromelia, bilateral clubfeet, small stomach	*FLNB*	Chr 3, 58062982, G→A	502G→A, Gly168Ser	NM_001164317.1	Missense	De novo	Atelosteogenesis (16752402)
fetal0178F	Femur <3% of normal size in third trimester	*FGFR3*	Chr 4, 1806119, G→A	1144G→A, Gly382Arg	NM_001163213.1	Missense	De novo (tier 2)	Achondroplasia (11030304)
fetal0183F	Lethal skeletal dysplasia, micromelia, abnormal profile, micrognathia, clubfeet, ambiguous genitalia, nuchal translucency of 3·5 mm	*COL2A1*	Chr 12, 48380868, C→A	1358G→T, Gly453Val	NM_001844.4	Missense	De novo	Atelosteogenesis (10797431)
fetal0173F	Unilateral renal agenesis, intrauterine growth restriction, lumbar hemivertebrae, complex cardiac defect	*KMT2D*	Chr 12, 49443635, TAG→T	3734_3735delCT, Ser1245TyrfsX4	NM_003482.3	Frameshift	De novo	Novel
fetal0129F	Enlarged polycystic and echogenic kidneys	*PKD2*	Chr 4, 88983135, ACT→A	2101_2102delTC, Ser701ArgfsX9	NM_000297.3	Frameshift	Maternal inherited	ADPKD (23300259)
fetal0045F†	Nuchal translucency of 3·5 mm	*RERE*	Chr 1, 8716108, C→CT	248dupA, Ser84ValfsX4	NM_001042681.1	Frameshift	De novo	Novel
fetal0164F	Hydrops	*RASA1*	Chr 5, 86674312, AAG→A	2446_2447delGA, Asp816LeufsX13	NM_002890.2	Frameshift	Maternal inherited	Novel
fetal0170F	Distended bladder, urethral atresia, enlarged stomach	*ACTG2*	Chr 2, 74141962, C→T	640C→T, Arg214Cys	NM_001199893.1	Missense	De novo	MMIH (24676022)
fetal0201F	Arthrogryposis, micrognathia, bilateral ventriculomegaly, polymicrogyria	*SCN2A*	Chr 2, 166201122, A→T	2620A→T, Ile874Phe	NM_001040143.1	Missense	De novo	Novel
fetal0210F	Abnormally shaped cerebellum, vermian hypoplasia, cardiomegaly	*COL4A1*	Chr 13, 110830552, C→G	2485G→C, Gly829Arg	NM_001845.5	Missense	De novo	Novel
fetal0222F	Hydrops	*SOS1*	Chr 2, 39251221, T→C	1132A→G, Thr378Ala	NM_005633.3	Missense	Paternal inherited	Noonan syndrome (21387466)
fetal0223F†	Dandy-Walker malformation, intrauterine growth restriction	*RAC1*	Chr 7, 6431617, A→C	170A→C, Asp57Ala	NM_018890.3	Missense	De novo	Novel
fetal0225F	Dandy-Walker malformation, vermian hypoplasia, meningocele	*CPLANE1*	Chr 5, 37165697, G→A	7477C→T, Arg2493X	NM_023073.3	Nonsense	Homozygous	Joubert syndrome (22425360)
fetal0238F§	Arthogryposis, micrognathia, pleural effusion, skin oedema	*RAPSN*	Chr 11, 47459593, A→G	1172T→C, Leu391Pro	NM_005055.4	Missense	Compound heterozygous	Novel

(Table 1 continues on next page)

	Prenatal imaging phenotype	Gene	DNA alteration*	cDNA alteration, protein alteration	RefSeq identifier	Consequence	Inheritance	Novel or previously reported (PMID)
(Continued from previous page)								
fetal0238F§	Arthogryposis, micrognathia, pleural effusion, skin oedema	*RAPSN*	Chr 11, 47469411, C→T	484G→A, Glu162Lys	NM_005055.4	Missense	Compound heterozygous	Congenital myasthenic syndrome (17594401)
fetal0239F	Several cardiac rhabdomyomas	*TSC2*	Chr 16, 2138293, CCGGCTCCGCCACATCAAG→C	5037_5054delC_A, His1679_Arg1684del	NM_001077183.2	In-frame deletion	De novo	Tuberous sclerosis (9829910)
fetal WC0001F	Nuchal translucency of 3·8 mm, pleural effusions	*RIT1*	Chr 1, 155874263, T→C	319A→G, Met107Val	NM_001256821.1	Missense	De novo	Noonan syndrome (27109146)

DNA variant data were sourced from Genome Reference Consortium Human Build 37 (GRCh37). PMID=PubMed identifier. Chr=chromosome. fsX#=frameshift variant that introduces a termination codon # codons downstream. NA=not applicable. ADPKD=autosomal dominant polycystic kidney disease. MMIH=megacystis microcolon intestinal hypoperistalsis syndrome. *Shown as chromosome, repeat identifier, change. †Initially considered bioinformatic signature and later changed to genetic diagnosis. ‡Splice change outside the protein-coding sequence. §Dizygotic twins.

Table 1: **Diagnostic genetic variants that were identified in a cohort of unselected fetuses with structural anomalies**

in fetuses with more than two anomalies (eight [35%] of 23 fetuses) compared with fetuses with fewer than three anomalies (16 [8%] of 211 fetuses; 6·4, 2·0–19·3; p=0·0007). The frequency of genetic diagnoses varied by the anatomical systems involved: the greatest proportions of diagnostic genetic variants were found in fetuses with structural anomalies to their lymphatic or effusion (24%), skeletal (24%), central nervous (22%), and renal (16%) systems, and the lowest proportion of diagnostic genetic variants were found in those with cardiac anomalies (5%; table 2).

The proportion of diagnostic variants found did not significantly differ between fetal DNA samples from cultured tissue (found in eight (10%) of 82 fetal DNA samples that were sequenced from culture) and those from direct samples (found in 16 (11%) of 152 fetal DNA samples that were sequenced from direct samples; Fisher's exact two-tailed test; odds ratio 0·9, 95% CI 0·3–2·4; p=1·00).

Four additional findings unrelated to the structural anomalies were identified (appendix). These variants included a well described homozygous nonsense Factor 11 pathogenic genotype in a fetus ascertained to have an umbilical hernia (RefSeq number NM_000128.3; gene *F11*; cDNA alteration 403G→T; protein change Glu135X); a de novo missense mutation in *MYL2* in a fetus with an atrioventricular canal (NM_000432.3; *MYL2*; 484G→A; Gly162Arg), which has been associated with hypertrophic cardiomyopathy;[20,21] a paternally transmitted *KCNQ1* variant (NM_000218.2; *KCNQ1*; 968G→A; Trp323X) and a maternally transmitted *PKP2* protein-truncating variant (NM_004572.3; *PKP2*; 1643delG; Gly548Val, with a frameshift that introduced a stop codon at codon 15).

	Number of fetuses*	Diagnostic genetic variants (% trios)	Bioinformatic signatures (% trios)	Number of bioinformatic signatures that are also a diagnostic variant†
Heart	77 (49)	4 (5%)	17 (22%)	4 (24%)
Nuchal	51 (32)	6 (12%)	10 (20%)	4 (40%)
Central nervous system	49 (29)	11 (22%)	16 (33%)	9 (56%)
Skeletal	34 (12)	8 (24%)	8 (24%)	6 (75%)
Intrauterine growth restriction	29 (5)	3 (10%)	5 (17%)	3 (60%)
Renal	25 (13)	4 (16%)	6 (24%)	1 (17%)
Lymphatic or effusion	21 (5)	5 (24%)	3 (14%)	2 (67%)
Other isolated anomalies‡	15 (15)	0	0	··

Fetuses with several anomalies are counted several times. *Number in brackets is the number of fetuses with a single structural anomaly in this anatomical system. †Frequencies reflect the proportion of diagnostic genetic variants that overlapped with a bioinformatic signature. ‡Abdominal wall (n=3), face (n=3), gastrointestinal tract (n=6), spine (n=1), and thorax (n=2).

Table 2: **Distribution of diagnostic genetic variants and bioinformatic signatures across the anatomical systems of unselected fetuses with structural anomalies**

Discussion

In a cohort of pregnancies with unselected fetal structural anomalies in which standard testing by karyotype and CMA were negative, we found that WES identified a well described genetic cause in 10% of cases. A genetic diagnosis was achieved 3 times more frequently in fetuses with several anomalies than in those with one anomaly (19% *vs* 6%). In fetuses with three or more anomalies, more than 30% had a diagnostic genetic variant.

The strength of our study was that it included all sequential fetuses with a structural anomaly rather than selecting cases with specific in utero phenotypes. Although this approach made our study more generalisable, it likely led to our lower prevalence of diagnostic genetic variants than the 25–35% reported in postnatal series.[3,9] These case series have a higher a priori risk of findings than our study because the phenotypes chosen for sequencing have often been meticulously evaluated by clinical geneticists and had additional imaging studies and detailed neurocognitive testing before sequencing. In our study, selection of fetal cases was challenged by restrictions to the phenotypes that can be identified in utero, including the inability to evaluate the fetal neurocognitive status, which is frequently the characteristic that prompts genetic sequencing in a child.

Confirmation that a variant identified in utero is causative and represents a genetic diagnosis is made

more difficult by our limited knowledge of in utero developmental phenotypes, which can differ from those described in the neonate and child. This limitation is illustrated in our case, fetal0045, in which a first-trimester ultrasound found an isolated enlarged nuchal translucency and WES revealed a hot zone essential de novo mutation in *RERE*; however, it was not until the fetus was older than 6 months (postnatally) that other related characteristics were seen that were consistent with a 2018 study[22] that implicated *RERE* in disease, and the variant was classified as pathogenic during re-assessment at a multidisciplinary team meeting. This difficulty also could have led to the high number of identified bioinformatic signatures that are not considered genetic diagnoses, some of which are likely causative but cannot be confirmed until similar cases are identified. Increased application of WES to fetal diagnostics and the routine reporting of well phenotyped cases to data repositories will lead to novel genotype–phenotype associations that are presently undocumented. For example, during our study, four fetuses that were initially considered to have bioinformatic signatures were reclassified as having diagnostic genetic variants on the basis of emerging medical literature that described a fetal phenotype that was not previously recognised in postnatal cases (table 1). The results from our bioinformatic signatures analyses highlight their value as a novel research-based approach to prioritise and study implicated variants in the absence of a confirmed genetic diagnosis.

The number of genetic diagnoses also varied by the anatomical system involved: we found the highest frequency of diagnostic genetic variants in fetuses with structural anomalies in the lymphatic (including effusions), skeletal, central nervous, and renal systems in our cohort (table 2); however, larger studies are required for greater precision in these frequencies. There are two potential explanations for the variability across anatomical systems. The first is that anomalies in certain organ systems might be more likely to be caused by a single gene variant than those in other systems. Alternatively, it could be that we have poorer knowledge of the in utero phenotypes associated with developmental genes in certain systems and are less able to confirm a genetic diagnosis. Until now, these explanations have been impossible to distinguish. However, since the detection of bioinformatic signatures are unbiased by our present knowledge of the potential phenotypes, their relative frequency can give insight into the proportion of anomalies that potentially have a detectable underlying genetic cause. For example, despite a difference in the frequency of diagnostic genetic variants between fetuses with skeletal anomalies and those with an enlarged nuchal translucency (24% *vs* 12%), both groups had relatively similar frequencies of bioinformatic signatures (24% *vs* 20%). This discrepancy in genetic diagnoses is primarily driven by fewer confirmed diagnostic genetic variants among fetuses with an isolated enlarged nuchal translucency in early gestation (appendix). For example, among 30 fetuses with an isolated nuchal translucency in the first trimester for which pregnancy outcome data were available, 29 (97%) fetuses were liveborn. By contrast, of 19 fetuses with nuchal translucency and additional anomalies, only five (26%) fetuses were liveborn; 12 (63%) of the pregnancies were terminated and two (11%) fetuses were stillborn (odds ratio 70·0 95% CI 7·9–3454; p<0·0001). It is likely that, as we gain more knowledge of the early gestational phenotypes of in utero syndromes associated with an enlarged nuchal translucency, more cases could prove to have a genetic basis.

Integration of WES into clinical prenatal care requires both an ability to identify the underlying cause of a disorder and evidence that care is improved. Previous studies[23,24] in critically ill neonates have suggested that establishment of a specific genetic diagnosis will lead to improved management and cost savings. Meng and colleagues[23] evaluated diagnostic WES that was used in the first 100 days of life in 278 critically ill neonates, and they identified a genetic cause in 36·7% of these neonates, which led to care modification in 52% of the cohort. The greatest effects were in infants who received rapid critical trio sequencing early in life[24] suggesting that sequencing early in the disease course could provide the maximal benefit and potentially improve clinical, developmental, and family psychosocial outcomes. This finding raises the important question of whether prenatal knowledge of a genetic diagnosis could further improve perinatal management by altering delivery plans, allowing caregivers to offer in utero treatment,[25] identifying damaging metabolic disorders before birth to prevent acute neonatal crises,[26] or allowing the parents the option of compassionate care when a genetic disease with a fatal long-term prognosis is detected.

Identifying a specific diagnostic genetic variant can facilitate parental reproductive counselling, and it can lead to improved management of future pregnancies by allowing prenatal or preimplantation genetic diagnosis. In our study, more than a third of the families (nine of 24 families) who were identified to have a diagnostic genetic variant had a risk of recurrence in a future pregnancy. Three of these pathogenic genetic variants were transmitted from a parent unaware of their genetic diagnosis (appendix). Of the remaining six fetuses with recessive disorders and carrier parents (those with mutations in *L1CAM*, *FLVCR2*, and *TMEM67*), the couple had at least one previously affected child before WES. Two of the carrier couples (those carrying *FLVCR2* and *C5ORF42* mutations) have already pursued chorionic villus sampling for specific variant identification in a subsequent pregnancy. One couple is currently pursuing preimplantation genetic diagnosis (those with a *RAPSN* mutation). Unfortunately, one family with an unreported bioinformatic signature (*ARMC9*) had a second affected pregnancy with the same phenotype before sufficient

information was available in the medical literature to make a genetic diagnosis. This outcome illustrates the delicate balance between overreporting and underreporting suspicious but uncertain results.

Our study confirms the feasibility and potential value of prenatal WES, but important logistical issues need to be addressed before widespread clinical implementation can be recommended. Consensus guidelines[27] from three major prenatal diagnostic organisations (the International Society of Prenatal Diagnosis, the Society of Maternal Fetal Medicine, and the Perinatal Quality Foundation) were published to outline these issues. A major emphasis of these guidelines is that interpretation of prenatal sequencing is complex and requires the availability and collaboration of a multidisciplinary team, including clinical and molecular geneticists, genetic counsellors, and fetal imaging specialists, to work together to obtain a diagnosis. The workflow must be optimised to minimise the turnaround time so that results are returned in time to allow families and physicians to use the information for pregnancy management. In our study, in which results were not intended to be used for clinical care, it took 4–8 weeks to obtain and interpret WES results. However, previous studies[28,29] have shown that results can be obtained within 72 h when necessary, and results can be obtained routinely within 2 weeks.

Some discretion was necessary in classifying a variant as a true diagnostic genetic variant because there is a paucity of reports describing genetic causes of early developmental phenotypes. For our study, we chose a conservative approach, in which we only clinically confirmed a genetic diagnosis when the study team judged the evidence to be sufficient for classification of the variant as pathogenic or likely pathogenic by the American College of Medical Genetics and Association for Molecular Pathology criteria for variant classification. Because the study team included clinical molecular geneticists, these judgements are likely to be broadly consistent with current clinical practice. This approach clearly underestimated the frequency of genetic causes, as illustrated by the fact that some of the bioinformatic signatures transitioned to be classified as diagnostic genetic variants during the study. With the development of fetal specific phenotype–genotype data repositories, uncertainty in cases should become less frequent.

In our study, WES of parent–fetus trios showed an underlying genetic cause in 10% of fetuses with structural anomalies that were negative in karyotype testing and CMA. The prevalence of WES-based diagnostic genetic variants differed by organ system involvement and these variants are more frequent in fetuses with several anomalies. Our experience with the application of diagnostic WES makes clear its clinical value while emphasising the challenges of establishing the appropriate multidisciplinary teams necessary to address the unique challenges of WES-based diagnostics in the prenatal setting.

Contributors

SP, VA, LB, NS, VJ, ZR, XZ, CMe, QW, SK, CMa, and BL analysed the data and SP, VA, DBG, and RJW interpreted the findings. SP generated the bioinformatics workflow and performed the statistical analyses. SP, VA, JLG, MS, LB, KB, NS, VJ, XZ, KA-Y, RM, BL, DBG, and RJW were members of the multidisciplinary clinical and genetics team. SP, VA, JLG, DBG, and RJW wrote the manuscript. JLG, MS, KW, ES, KB, KA-Y, and ON and RM collected data. LB assisted with the development of the protocol, CMa performed the sequencing, and BL performed the microarrays. DBG and RJW designed the study.

Declaration of interests

SP reports employment with AstraZeneca. All other authors declare no competing interests.

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